Competency Level 9.12: Manages data in databases

Time: 04 periods

Learning Outcomes:

- Embeds SQL statements in programming languages to retrieve, add, modify and delete data

Contents:

- Connecting to a database
- Retrieve data
- Add, modify and delete data

Concepts and terms to be highlighted:

- Appreciate the fact that programs need to get data and information from saved database file
- Appreciate how to create a simple database to do database handling operations
- Appreciate how to refer to a database in a program to get connected to a database
- Appreciate how to open a database and how to create a suitable interface program containing forms with validations to retrieve, update, delete and search a database record conveniently
- Appreciate the application of SQL statements to do the above mentioned operations
- Appreciate the simple multiple tables applications of databases to solve practical problems
- Database connectivity, Database handling operations

Guidance for lesson plans:

- Divide the students in to groups
- Ask the groups to create a suitable simple database
- Ask the groups to write a program to get connected to a database
- Ask students to create a program for a given scenario (database) with validations to retrieve, update, delete and search a database record conveniently

Guidance for assessments and evaluations:

- Ask the groups to create a suitable simple database for a given scenario
- Ask the groups to write a program to get connected to the developed database
- Ask students to create a program for a given scenario (database) with validations to retrieve, update, delete and search a database record conveniently

Quality inputs:

- Computers with python programming language and database software
- Sample programs
- Presentations

Competency Level 9.13: Searches and sorts data

Time: 04 periods

Learning Outcomes:

- Uses sequential searching technique appropriately
- Implements bubble sort technique appropriately

Contents:

- Searching techniques
    - Sequential search
- Sorting techniques
    - Bubble sort

Concepts and terms to be highlighted:

- Appreciate the fact that in real life situations, we have to sort things in order to search and find them fast and easily
- Appreciate the need for sorting data in order to search and find them fast and easily
- Appreciate the concept of searching using Sequential Search techniques in a data set stored in an array
- Appreciate the concept of sorting using Bubble Sort technique
- Appreciate the concept and technique (the need for a dummy variable) of swapping two data items
- Sorting and Searching, Sequential Search, Bubble Sort

Guidance for lesson plans:

- Give numbers randomly to a set of students in the class
- Ask a student to find the student holding a specific number and let the students feel the difficulty in finding the number due to the fact that the numbers are not sorted
- Ask the groups to write a program according to the physical method they were using for searching (Sequential Search)
- Ask students to stand in to a row in an order of the number (ascending or descending) using the Bubble Sort technique starting from the original unsorted set
- Ask the groups to write a program according to the physical method they were using for sorting (Bubble Sort)

Guidance for assessments and evaluations:

- Ask the students to write a program to search a data according to a given situation using Sequential Search
- Ask the students to write a program to sort data according to a given situation using Bubble Sort

Quality inputs:

- Computers with python programming language
- Sample programs
- Presentations

Competency 10: Develops websites incorporating multi-media technologies (using HTML 5)

Competency Level 10.1: Explores the need for web

Time: 08 periods

Learning Outcomes:

- Describes www
- Analyses the systematic arrangements of contents and structure of a web

Contents:

- The world wide web (www)
- Types of web sites
    - Information, news
    - Personal, educational, commercial, Research
    - Web portals

Concepts and terms to be highlighted:

- Appreciate the need for information sharing and communication with others.
- Appreciate the important of computer network in information sharing and communication
- Appreciate the important of the internet and the www in as a global network of computers for information sharing communication.
- Internet
- www
- web portals
- types of web sites

Guidance for lesson plans:

- discuss the need for the web
- discuss computer network as the base for the web
- discuss the www and evolution of www
- divide the student into group
- ask each group to browse the internet search information for given topics the teacher
- Request group to categories the web sites
    - Ex: educational, personal, commercial, etc.

Guidance for assessments and evaluations:

- Ask student to visit website given by the teacher
- Ask student to visit search engine (Google, yahoo, etc.)
- Ask student to search particular topics using to search engine
- Ask them to visit some of the search results (sites)
- Ask student to find the most appropriate site among the search result.
- Ask the student to categories the web site according to given topics (educational, personal, commercial, etc.)

Quality inputs:

- Internet access facility ,Presentations

Competency Level 10.2: Analyses user requirements (multimedia contents)

Time: 04 periods

Learning Outcomes:

- Creates effective and appropriate information layout of a website
- Identifies the web pages of a website
- Identifies the contents of a web page
- Identifies navigation structure

Contents:

- Defining the objectives of a website
- Contents to be displayed

Concepts and terms to be highlighted:

- To appreciate the need of web site to transmit information effectively
    - Ex: commercial web site is a very useful and effective marketing tool
        - Educational web site as a very useful and effective teaching tool
- To appreciate why it is necessary to identify the target groups of web sites.
    - Ex:- customers of a bank
        - Student in a school
- To identify the most useful and effective information in the web site according to the target groups.

Guidance for lesson plans:

- Divide the student into groups
- Provide web address to each group and ask them to visit those website
- Ask the student to identify the objective of the website
- Discuss the purpose and an objective of each websites
- Ask the groups to observe the website and write the contents displayed
    - Ex: animation, list, headings, pictures, videos, hyperlinks, etc.
- Discuss the content of website and its organization structure

Guidance for assessments and evaluations:

- Divide student into groups
- Provide deferent web address to each group and ask them to visit that website and identify the objective of the web sites and present it.
- Ask the groups to observe the website and identify the structure and contents to displaying and present it.

Quality inputs:

- Internet facility, print out with web address, presentations

Competency Level 10.3: Identifies appropriate HTML tags to design a single web page

Time: 04 periods

Learning Outcomes:

- Analyses the arrangement of contents of a web page
- Analyses the organization of contents in a web page
- Creates a simple web page

Contents:

- Building blocks of a web page
  - Page definition
  - <html></html>
  - Head section
  - <head></head>
  - <title></title>
  - Body section
  - <body></body>
- Background color
  - Text formatting
  - <h1>…<h6>tags
  -
  -
  - Underline, bold, italic
  - <font></font>
  - Size and color
- Adding comments

Concepts and terms to be highlighted:

- Appreciate that webpages are developed using HTML
- Appreciate that HTML provide various tags to structure the web page
- HTML
- Tags
- Comment
- Attributes
- Elements
- Text editor (notepad, notepad ++)

Guidance for lesson plans:

- Divide the student into group
- Provide teacher developed simple web page to each group and ask the groups to observe the page source and make some miner editing and observe the corresponding out puts.
- Discuss and introduce the building blocks of web page and tags.
  - Page definition
  - Head section
  - Body section

  - Formatting
  - Adding comments etc.
- Ask student groups to construct simple webpage using a text editor. (notepad is recommended)

Guidance for assessments and evaluations:

- Provide the requirements of the web page and ask the student to develop an appropriate simple web page
- Ex:- to develop a simple web page for student profile- name, address, education qualification etc. with his/her photograph

Quality inputs:

- Computers with simple text editors, web browser, presentations

Competency Level 10.4: Uses HTML to create linked web pages

Time: 16 periods

Learning Outcomes:

- Explains hypertext markup language
- Identifies the standards of HTML
- Saves the source document with suitable extensions
- Designs the web page by inserting appropriate multimedia objects according to user requirements
- Organizes data using lists and tables in the web page
- Links pages and multimedia objects to the web page

Contents:

- Contents of a website
  - Home page
  - Linked pages
  - Hyperlink
    - ❖ Different sections of the same page(book mark)
    - ❖ Different pages of a same site(local link)
    - ❖ Pages of different sites (External link)
- Lists
  - Ordered lists
  - Unordered lists
  - Definition lists
- Image
- Tables
  - <table></table>
  - <th></th>
  - <tr></tr>
  - <td></td>
  - <caption>
  - Merging columns and rows
- Multimedia objects
  - Audio
  - Video

Concepts and terms to be highlighted:

- To appreciate the structure of a composite website containing several webpages including list, image, table, etc. connected in a logical manner.
- To appreciate the purpose of dividing into several logical pages and why and how there are connected using a hyperlink
- Home page, link page, hyperlink
- List, image, table
- Multimedia objects (audio, video, animation)

Guidance for lesson plans:

- Divide student into group
- Ask the group to visit the suitable website given by the teacher and identify the different component (homepage, link page, list, hyperlink, table, multimedia contents, etc) in the site.
- Discuss and describe each of component and their constructions.
- Discuss the proper save methods with extension (.html, etc)

Guidance for assessments and evaluations:

- Provide the partially completed page source and ask the student to complete appropriately to given web page(filling)
- Provide requirements for a website and ask the students to develop a website accordingly (should include home page, hyperlinks, list, table, multimedia contents)

Quality inputs:

- Computers with simple text editors, web browser, presentations
- Requirement description

Competency Level 10.5: Uses Style sheet to change the appearance of web pages

Time: 04 periods

Learning Outcomes:

- Briefly explains style sheet and its usage
- Uses the comments and correct syntax in CSS
- Uses appropriate selectors to select elements in CSS
- Inserts CSS in HTML web pages to improve the appearance
- Applies various CSS formatting in HTML web pages to improve the appearance

Contents:

- Introduction to style sheet
- CSS
  - Syntax, comments
- CSS selectors
  - element, id, class, group
- Ways of inserting CSS
  - Internal, external, inline
- Appearance formatting
  - Background (color, image)
  - Text and fonts
  - Links
  - Lists
- Tables

Concepts and terms to be highlighted:

- Appreciate the importance of appearance of a web page
- Cascade style sheet (CSS)
- Appreciate that CSS is an improvement of HTML

Guidance for lesson plans:

- Divide the student into groups
- Provide teacher developed two web pages, one CSS based and other without CSS.
- Ask the groups to compare the two web pages and their corresponding source codes and present their findings
- Discuss the importance of appearance (attractiveness, user friendliness and creativity) of web page
- Introduce the cascade style sheet
- Discuss and describe the construction of CSS based web pages
- Provide few web pages developed using HTML only and ask student groups to improve them using CSS

Guidance for assessments and evaluations:

- Provide partially done pages using CSS and ask the groups to complete them